

SAINTS Year 1 Progression Document

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1 Provisional Title of Thesis

World Models for Prediction in Safety Critical Applications

2 Brief Description of the Research Topic

Embodied agents must act from incomplete, high dimensional observations while anticipating the consequences of their actions. This is particularly difficult for legged robots in previously unseen indoor environments. A single egocentric image may support immediate collision avoidance, but cannot reliably reveal which corridors have already been explored, whether a visually repetitive junction is new, or how to reach a goal that is outside the current field of view. The resulting problem combines continuous control, partial observability, perceptual aliasing, long horizon planning, and direct physical safety consequences.

This research investigates whether compact predictive world models can provide a useful substrate for this problem. The central model family is the Joint Embedding Predictive Architecture (JEPA), which predicts future representations rather than reconstructing future pixels. An observation is encoded into a latent state and an action conditioned predictor estimates how that state changes under candidate controls. Planning can then operate in a compact representation space, potentially avoiding the cost and irrelevant detail of pixel level generative rollouts. The research asks whether this representation preserves the geometry, semantics, and controllability needed for embodied planning, rather than treating low prediction loss as sufficient evidence.

The current draft paper, *An Action-Conditioned JEPA with Online Topological Memory for Quadruped Navigation*, studies the complementary roles of local prediction and persistent memory, and their efficacy in navigating unseen mazes, with the goal to collect hidden beacons. A short horizon action conditioned JEPA can estimate the local consequences of candidate action sequences and support traversability, collision risk, and exploration costs. It cannot by itself retain a reliable account of places visited beyond its context and rollout horizon. An online topological memory therefore stores place like latent nodes, action conditioned transitions, frontier information, and beacon adjacent observations. Under perceptual aliasing, the intended memory state is a belief distribution over plausible nodes, updated using observations, executed actions, graph consistency, and optional proprioceptive evidence, rather than an immediate commitment to one location.

This division of labour is the central research claim. The learned world model handles local dynamics and immediate action consequences, persistent memory handles long horizon connectivity, revisitation, and ambiguity over place. Topological memory and latent planning are not novel in isolation. The contribution is to investigate their interface. Do action conditioned JEPA states form a useful substrate for online memory and local predictive planning, do rollouts add value beyond reactive learned costs, and does explicit memory over JEPA states improve navigation under task inherent aliasing?

The target system is an RGB only high level navigation stack for a Unitree Go2. Egocentric observations are encoded into a learned spatial state, currently a 64×64 bird's eye view (BEV) representation

decoded from ViT tokens by learned queries. An action conditioned predictor estimates future states, and a planner combines local predicted costs with a persistent place graph. A conventional low level locomotion controller remains beneath the stack, the thesis contribution is high level predictive navigation rather than gait learning.

Year 1 work progressed from a VICReg latent MPC pilot in *CarRacing-v3*, through recurrent and EMA target quadruped JEPAs, to a ViT/Transformer *LeWorldModel* implementation using SIGReg. These systems established that compact action conditioned representations can learn a scene dependent structure and support limited local planning. They also exposed structural and methodological limits: camera views were corrupted by geometry clipping in an early simulator pipeline, scene level splitting and reset hygiene were essential, safe local control did not produce adequate exploration, and hybrid classical components could mask weakness in the learned model.

The current *LeWMQuad-v3* repository therefore applies fixed training, selection, calibration, untouched, and sealed data roles, preregistered gates, immutable attempt identities, and component specific baselines. The active bottleneck is the RGB to spatial representation. The original ray/depth/raster model fit a small training family of mazes, but failed on held out maze families. Secondly, a shared cap on the size of each training update meant that the much stronger camera learning signal consumed almost the entire update, leaving the JEPA objective (responsible for modelling the consequences of the robot's actions) with virtually no influence, an imbalance of approximately eighty four million to one. Direct learned BEV variants show RGB dependence and useful geometry, but no checkpoint has yet preserved occupied space semantics strongly enough to pass the complete physical gate.

If perception qualifies, the next stage is genuine joint JEPA learning. An online encoder, spatial representation, and action conditioned predictor will receive predictive gradients against a stop gradient EMA (Exponential Moving Average is a slowly updated copy of the encoder that provides a stable reference and is not directly trained by the prediction error) target. A semantic auxiliary objective will remain active, with its gradients constrained independently from the JEPA gradients so that neither objective can dominate the shared representation. Preregistered non regression criteria will then reject any checkpoint that gains predictive capability at the expense of the spatial semantics established during perception qualification. We will subsequently evaluate local rollout and online memory through controlled ablations and one way held out testing.

The later research contribution aim is a hierarchical extension that separates short horizon dynamics from slower strategic transitions between places or sub goals. It will be tested against the strongest flat JEPA plus memory baseline, rather than assumed to help. The intended research arc is first to qualify a physically useful predictive state, then establish the contribution of local rollout and persistent memory, test multi timescale prediction, evaluate transfer to the physical Go2, and finally to develop measurable prediction, memory, and planning artefacts for safety arguments.

3 Research Questions

These research questions are motivated jointly by the Year 1 experimental and the research gaps established in the literature review below. Together, they identify unresolved questions around transferable spatial representation, predictive planning, persistent memory, temporal hierarchy, and safety evidence.

1. **RQ1: Predictive representation.** Can a minimal sensor based learned spatial representation and action conditioned JEPA be trained to preserve the geometry and semantics required for navigation, generalise across novel environments, and support accurate short horizon prediction without relying on privileged non learned map evidence at inference?
2. **RQ2: Local rollout and persistent memory.** Do action conditioned latent rollout and online topological memory provide complementary benefits for exploration and goal based navigation under partial observability and perceptual aliasing?
3. **RQ3: Multi timescale extension and transfer.** Once the flat action conditioned JEPA plus persistent memory stack is established, does a hierarchical JEPA with separate prediction horizons improve long horizon planning efficiency, robustness, and transfer to a physical Unitree Go2?
4. **RQ4: Safety evidence and runtime conservatism.** Which measurable signals from predictive representation, memory, and planning can support runtime intervention and a structured safety argument for learned quadruped navigation?

A negative result remains informative. Examples include showing that a semantic auxiliary cannot preserve occupied space during joint JEPA training, that reactive heads perform as well as latent rollout, that graph memory over pretrained features matches graph memory over JEPA states, that a full place belief adds no value over a single node, or that a hierarchical predictor does not improve over a properly controlled flat baseline. The experimental protocol is designed so that these outcomes can be reported without reinterpretation of the held out benchmark.

4 How the Research Relates to Safety

The safety contribution is not the claim that a JEPA is intrinsically safe, nor that low collision counts alone demonstrate acceptable behaviour. Safety requirements for the system must address both safe operation and the consequences of failing to perform its intended task. For a quadruped used in search or inspection, these may include avoiding collisions, falls, and property damage, alongside minimum progress, coverage, and task completion requirements where failure could leave a hazard undetected or an area uninspected. Evaluation must therefore consider operational safety and safety related task performance together. A stationary robot may satisfy a collision avoidance requirement while still violating the wider safety requirements of the task.

The current development gates provide one concrete safety mechanism. A model cannot enter navigation evaluation unless its learned representation passes preregistered occupied space, free space, and physical motion criteria. An oracle has shown that the gate is achievable, and learned candidates that fail are blocked rather than promoted as the “closest” result or tuned on untouched scenes. This creates a traceable link from a physical requirement to immutable evidence.

The proposed system also separates visual representation, action conditioned prediction, local risk and task costs, place memory construction, belief update, graph planning, and low level locomotion. This does not guarantee safety, but it permits more precise failure attribution than a monolithic policy. The earlier gradient analysis illustrates the value. Apparent predictor weakness was partly caused by an auxiliary camera objective overwhelming JEPA gradients, while classical guards could suppress collisions without demonstrating learned navigation competence.

The research will evaluate candidate runtime monitors only against labelled events and calibrated operating conditions. These include collision, clearance, and traversability estimates, loss of occupied space recall, and distribution shift surprise. Measures will include precision recall, calibration error, and the task cost of conservative intervention. A monitor is useful only if it detects risk early enough to alter execution without making the system functionally inert.

The intended output is a structured evidence package rather than a complete assurance claim. Candidate claims concern perception qualification, bounded prediction error over the planning horizon, collision risk detection before contact, belief behaviour under perceptual aliasing, minimum coverage and task progress, and conservative fallback outside the calibrated envelope. Supporting artefacts will include gate receipts, per seed results, prediction and belief traces, failure mode maps, ablations, and physical replay logs.

Physical Go2 experiments will be staged at conservative speed in a controlled area with padded obstacles, and a human operator. Learned navigation will not be transferred until the corresponding simulator gates pass. Raw observations, commands, predicted risk, memory state, belief state, and intervention events will be logged for offline reconstruction.

Crucially, the ability of the system to produce multiple predicted futures based on candidate actions, and for those predicted futures to be evaluated based on the above criteria, allows for tuning the system to the appropriate risk tolerance for its intended deployment context.

5 Plan for Year 2

Year 2 runs from October 2026 to September 2027. The plan is dependency driven, perception qualification precedes joint JEPA training, qualified prediction precedes claims about latent rollout, and the complete Paper A stack is frozen before one way untouched or sealed evaluation and before learned physical navigation. Preparatory implementation may overlap between work packages, but no dependent experiment or claim will proceed until the preceding milestone has been met.

The work packages map directly onto the research questions. WP1 and WP2 address RQ1 by establishing a transferable RGB derived spatial representation and determining whether it can support action-conditioned prediction. WP3 addresses RQ2 by testing whether short horizon predictive rollout and persistent memory provide complementary benefits. WP4 addresses the multi-timescale hypothesis in RQ3, while WP5 provides the corresponding physical-transfer evidence. RQ4 is supported by all work packages, they progressively evaluate whether failures or uncertainty in perception, prediction, memory, and planning can support runtime intervention and a structured safety argument.

WP1: RGB to BEV Perception Qualification (RQ1 + evidence for RQ4)

October–December 2026. The aim of this work package is to establish an RGB-only, map-like bird's-eye-view representation that reliably preserves the geometry and obstacle semantics required for navigation in previously unseen mazes. This capability must be demonstrated before later prediction or navigation results can be interpreted reliably.

The current representation will be evaluated against fixed, preregistered perception criteria. If it does not qualify, the resulting failure evidence will be used to select one substantively different successor approach while retaining the fixed data roles and RGB only constraint.

Milestone M1: Either a qualified perception checkpoint that can initialise joint JEPA training, or a documented negative result with a justified alternative representation direction.

WP2: Joint JEPA Training (RQ1 + enabling RQ2)

November 2026–February 2027. Preparatory implementation and baseline testing may begin during WP1, but joint training will begin only after perception qualification. The aim is to determine whether the learned spatial representation can predict how the robot's nearby environment will change following candidate actions, while retaining the free space and obstacle information established in WP1.

The encoder, spatial decoder, and action-conditioned predictor will be trained using a slowly updated reference encoder that provides a stable prediction target. The semantic and predictive training signals will be monitored and controlled separately so that neither can overwhelm the other. Evaluation will distinguish accurate single step prediction from useful multi-step rollout, where errors can accumulate, and will test whether the model responds appropriately to different candidate actions. Comparisons with a model without action conditioning and a reactive controller without predictive rollout will establish whether prediction adds value beyond current-state costs.

Milestone M2: A jointly trained representation and predictor that retain the qualified spatial semantics

and pass the predefined prediction and rollout criteria.

WP3: Online Memory and Paper A (RQ2 + evidence for RQ4)

January–June 2027. Memory interfaces and evaluation infrastructure can be prepared in parallel with WP2, but claims about memory and rollout will depend on M2. The aim is to test the proposed division of labour between local prediction and persistent memory. The predictive model will estimate the immediate consequences of candidate actions, while an online place graph will retain information about previously visited locations, transitions between them, unexplored regions, and observed beacons.

Because visually similar corridors may correspond to different physical locations, the memory will maintain confidence across multiple plausible locations rather than always committing immediately to one. The evaluation will compare retaining a full distribution over possible locations, a short list of likely locations, and a single best estimate. It will measure incorrect place merges, false loop closures, unresolved localisation uncertainty, recovery after an incorrect hypothesis, and convergence after informative actions.

Controlled comparisons will isolate the contributions of action conditioning, predictive rollout, learned costs, persistent memory, the underlying visual representation, and multi-hypothesis localisation. Prediction, memory, and complete navigation performance will be reported separately using measures including coverage, exploration before a beacon is observed, beacon capture, task completion, collisions, path efficiency, and planning latency.

Milestone M3: A complete Paper A evaluation sufficient to answer RQ2, followed by one-way untouched evaluation after the code, checkpoint, thresholds, calibration, and environment versions have been frozen. The sealed V4 benchmark will be opened only after this complete freeze.

WP4: H-JEPA Extension (RQ3)

April–September 2027. Initial design and prototyping may begin while Paper A is being completed, but the controlled comparison will use the strongest qualified flat JEPA plus memory system from WP3 as its baseline. The aim is to test whether prediction at more than one timescale improves long-horizon navigation.

A tactical predictor will model short action sequences and their immediate geometric consequences, while a strategic predictor will model slower transitions between places, frontiers, or intermediate goals. The memory representation will be held constant where possible so that any improvement can be attributed to the additional prediction timescale rather than to introducing hierarchy and memory simultaneously. Evaluation will consider effective planning horizon, exploration efficiency, replan stability, robustness, computational cost, and transfer. A result showing no benefit over the flat baseline will remain informative.

Milestone M4: A minimal H-JEPA prototype and a controlled result showing whether multiple prediction timescales improve long-horizon navigation beyond a properly controlled flat baseline.

WP5: ROS 2 and Physical Unitree Go2 Pilot (RQ3 and RQ4)

February–September 2027. Software integration and scripted tests may begin before Paper A is

complete, but learned physical navigation will begin only after the corresponding simulation gates and M3 freeze have been passed. The aim is to examine transfer from simulation to the physical Unitree Go2 and to evaluate whether the proposed runtime warning signals remain useful under real sensing, timing, and motion conditions.

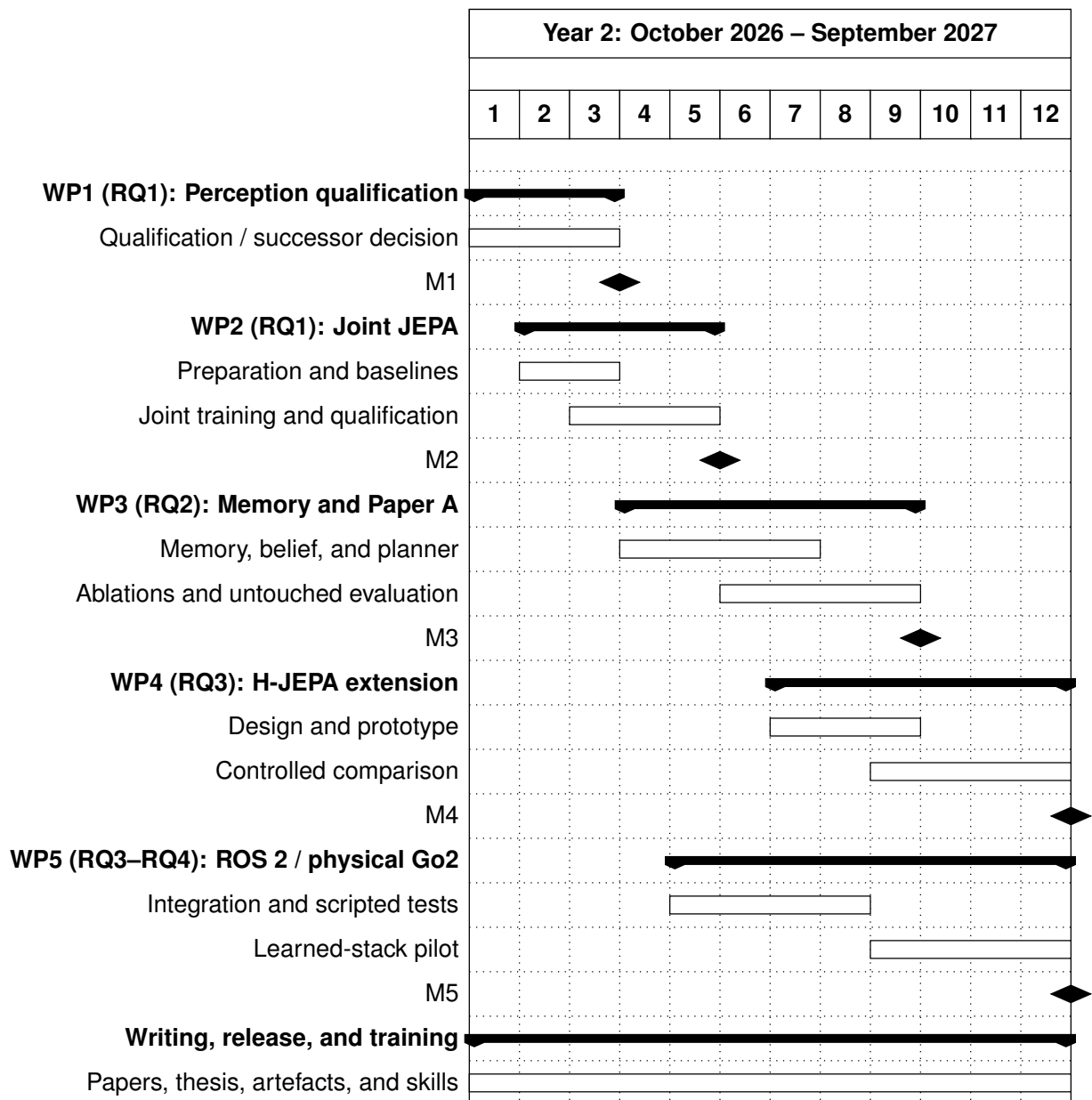
The qualified stack will be packaged as modular ROS 2 components. Scripted motion will first validate the camera interface, coordinate transforms, command timing, logging, and emergency stop behaviour. Low-speed learned navigation pilots will then examine transfer failures, runtime latency, prediction errors, localisation uncertainty, interventions, and the balance between conservative behaviour and task progress. Statistically powered comparisons will remain in simulation, the physical study will support bounded conclusions about transfer and runtime monitoring.

Milestone M5: A reproducible physical Unitree Go2 pilot with complete replay logs, documented transfer failures, and bounded conclusions relevant to RQ3 and RQ4.

Writing, Release, and Training

Writing and artefact preparation will run throughout the year and will consolidate evidence across all four research questions. Paper A hypotheses, methods, primary comparisons, and evaluation criteria will be frozen before untouched testing. The release will include split manifests, environment-generation code, configuration hashes, checkpoint metadata, and a reduced reproduction path. Training priorities are ROS 2 and safe Unitree Go2 operation, probabilistic localisation under perceptual aliasing, statistical design for stochastic robotics, latent geometry and state abstraction, multi-timescale predictive modelling, and the construction of structured safety arguments.

Gantt Overview



Month 1 is October 2026 and month 12 is September 2027. The detailed risk register is provided in Appendix C.

6 Detailed Risk Register

Risk	Likelihood	Impact	Mitigation and decision rule
Perception never passes the complete physical gate	Medium	High	Preserve fixed roles and thresholds. Close repeated minor variants. Use the failure signature to select one materially different representation mechanism. The negative result remains useful evidence about representation requirements.
Auxiliary semantics again starve JEPA learning	Medium	High	Measure raw and post control gradient norms and cosine similarity, use separate parameter groups and clipping, stop early if effective JEPA contribution collapses.
Latent rollout adds no value over reactive heads	Medium	Medium	Treat the reactive baseline as mandatory. If no benefit is observed, narrow Paper A to the representation memory interface and report rollout as a negative result.
Topological memory is dominated by hand set thresholds	Medium	Medium	Identify algorithmic choices, perform sensitivity analysis, compare learned and thresholded place or edge models where feasible, and avoid an end to end learned claim unless supported.
Aliasing causes false merges or loop closures	High	High	Maintain multi hypothesis belief, use transition and graph consistency, report false merges/closures, and trigger conservative behaviour while belief remains high entropy.
Planning cost grows with memory	Medium	Medium	Log update and search latency separately, use a preregistered pruning or compression policy, compare full and top- k belief.
Pressure to tune on sealed data	Low	High	Maintain immutable roles, one way access, attempt receipts, and an explicit freeze of code, model, thresholds, and environment before opening V4.
Sim to real performance is poor	High	Medium	Keep powered ablations in simulation, stage physical tests, categorise photometric, dynamics, latency, and interface failures, report a bounded transfer study.

Risk	Likelihood	Impact	Mitigation and decision rule
Year 2 scope becomes too broad	Medium	High	Paper A is the priority. H-JEPA and physical deployment are gated extensions and do not begin at scale before a qualified flat plus memory baseline exists.

7 Approach to Responsible Research and Responsible AI

The project uses the AREA framework (Anticipate, Reflect, Engage, and Act) as an iterative approach to responsible research and innovation [1]. The framework will be revisited at major experimental and deployment decisions rather than applied only retrospectively. Detailed experimental controls are specified subsequently, the emphasis here is on the wider consequences of successful research, the assumptions and values shaping the work, the stakeholders who should influence it, and the actions taken in response.

7.1 Anticipate

7.1.1 Potential Benefits and Application Boundaries

The research aims to produce general evidence about predictive representations, persistent memory, planning, and runtime monitoring for embodied autonomous systems. The current empirical case study is deliberately narrower. RGB based, high level navigation by a Unitree Go2 in bounded indoor environments. If successful, the work could support search and inspection in hazardous settings and provide more inspectable evidence about component failures. It is not a deployment ready product, and other application domains will require their own requirements, evidence, and stakeholder engagement.

7.1.2 Potential Harms, Bias, Privacy, and Sustainability

Operational harms include physical damage, unsafe motion, and failure to complete a safety related task. Excessive conservatism is also a concern. A stationary robot may avoid collisions while failing the wider safety requirements of its task. Simulation performance and low collision counts will therefore not be presented as safety guarantees.

Surveillance, human identification, target pursuit, and weaponisation are outside the intended scope. Simulator generated data contain no personal information, but may not represent all relevant conditions, so results will be reported across scene families and physical settings. Physical experiments will avoid recording uninvolved people, and incidental recordings will be minimised, access controlled, and excluded from public release. Gate based development and compact models will reduce unnecessary computation.

7.2 Reflect

7.2.1 Methodological Assumptions and Research Integrity

Synthetic mazes, RGB only perception, a bird's eye view representation, and a conventional low level controller simplify the problem but limit the claims that can be made. Results will therefore be bounded to the tested operating conditions.

Training, checkpoint selection, calibration, untouched, and sealed populations are kept distinct. Final mazes do not guide architectures, thresholds, seeds, schedules, or stopping points. Attempts and decision criteria are defined before execution, and operational failures that perform no scientific work are audited rather than silently rerun. These controls respond to Year 1 failures in data separation, reset hygiene, rollout evaluation, and component attribution. Scene level separation, episode safe sampling, rollout diagnostics, component specific baselines, and retention of negative results are therefore required.

7.2.2 Reproducibility, Uncertainty, and Disciplinary Limits

Exploratory engineering, preregistered experiments, and final evaluation will be distinguished explicitly, with deviations recorded before interpretation. The intended release includes source code, generation and evaluation procedures, checkpoints, source hashes, model identities, and a reduced reproduction path. Advice will be sought where conclusions require expertise outside machine learning.

7.3 Engage

7.3.1 Academic and Multidisciplinary Engagement

Supervisory review, SAINTS activities, and engagement across robotics, machine learning, safety assurance, statistics, and responsible innovation will challenge the experimental design and interpretation. Particular attention will be given to whether technical metrics correspond to meaningful physical and task requirements.

If the proposed healthcare facing supervision arrangement is confirmed, it will provide a route to appropriate healthcare stakeholders. No healthcare specific benefit or safety claim will be made without supporting engagement and evidence.

7.3.2 Industrial, Sectoral, and Public Engagement

DNV is a significant stakeholder for the maritime case study context. Engagement with DNV will examine operational hazards, the consequences of incomplete inspection, and the evidence meaningful to the maritime community and an assurance process. The stakeholder group will be expanded when a new application introduces materially different requirements. Research communication will acknowledge uncertainty, negative findings, operating limits, and the distinction between a research prototype and an assured autonomous system.

7.4 Act

7.4.1 Responsive Research Design and Safeguards

Evidence and stakeholder input will inform requirements, evaluation criteria, and design decisions. Recognition that non performance can have safety consequences has already led to evaluation of task progress alongside collision avoidance. Runtime monitoring will also be judged by the task cost of unnecessary intervention.

The project may change direction at explicit decision points in response to evidence or stakeholder concerns. Changes will be documented before later results are interpreted and will not justify retrospective alteration of frozen hypotheses, thresholds, or held out evaluations. Physical testing will proceed only after simulator gates pass, using conservative speeds, padded or lightweight obstacles, a controlled area, and a human safety operator. Complete traces will support reconstruction of incidents and bounded transfer claims.

7.4.2 Open Science, Compute, and Continuing AREA Review

Released artefacts will state the intended operating context, known limitations, and the distinction between research evidence and deployment assurance. Reproducibility will be supported through an open source GitHub repository and usage guide, subject to privacy, licensing, security, and partner constraints. Failed mechanisms will be closed rather than repeatedly rerun with cosmetic changes. AREA will be revisited at major milestones and when entering a new application domain, with resulting actions recorded.

8 Literature Review

8.1 Introduction and Scope

This review positions the thesis at the intersection of predictive world models, visual navigation under partial observability, persistent spatial memory, model based planning, and safety evidence for embodied AI. The central problem is not simply to predict the next observation or to build a map. It is to learn a compact state that supports controllable prediction, maintain information that cannot be recovered from the current frame, and use both for short and long horizon action selection on a physical platform.

The literature suggests three relevant abstractions. First, recurrent latent world models compress history into a learned belief state and support planning or policy learning through imagined rollouts. Second, feature predictive and JEPA style models avoid pixel reconstruction and plan directly in representation space. Third, neural and semi parametric navigation systems maintain explicit memories or graphs of places. The thesis tests a particular combination of a local action conditioned JEPA rollout for continuous action consequences, coupled to persistent online topological memory for place identity and long horizon connectivity.

8.2 Navigation as a Partially Observable Problem

A navigation agent does not observe the full environment state. Its camera provides a local projection conditioned on viewpoint, lighting, occlusion, and sensor noise. Distinct states can therefore generate similar or identical observations. In the POMDP formulation, an agent acts from a belief over latent states rather than a fully observed Markov state [2]. Perceptual aliasing is particularly severe in repeated corridors and junctions: a feedforward encoder can produce a useful local representation while still being unable to identify which instance of a corridor is currently observed.

This distinction is important for evaluating representation learning. A low place classification score does not necessarily show that the encoder has failed to learn useful local geometry as the observation itself may be insufficient. Conversely, strong local collision prediction does not show that the model can navigate to an occluded goal. Persistent memory and action history are needed to transform ambiguous local evidence into a belief over place. The thesis therefore separates observation representation, transition prediction, place belief, and graph connectivity rather than asking one latent vector to serve all roles without diagnosis.

State abstraction theory provides a useful vocabulary for this separation. An effective abstraction should preserve distinctions that affect future decisions while discarding irrelevant detail. Bisimulation inspired abstractions group states according to their task relevant outcomes and action conditioned transition behaviour, rather than visual similarity alone [3]. Here, this is used as a design principle rather than as a claim that the model is trained with a formal bisimulation objective. The local JEPA state should preserve nearby geometry, traversability, and the short horizon consequences of candidate actions. The persistent place representation should instead abstract away nuisance appearance variation while retaining place discriminative landmarks, reachability, and topological

connectivity. A later high level layer in the H-JEPA may operate over a coarser representation of this kind, but remains a separate controlled extension.

8.3 Generative and Latent World Models

World models learn a predictive model of environment dynamics and use it for control, planning, or policy improvement. Pixel generative approaches offer interpretable imagined observations but must allocate capacity to texture, illumination, and other details that may be irrelevant to action. Recurrent state space models instead learn compact stochastic latent dynamics and have demonstrated strong performance across diverse control domains, including the Dreamer family [4]. Their recurrent state functions as a learned belief, but planning and policy learning can still suffer from compounding model error and limited temporal abstraction.

Navigation World Models provides a navigation specific generative comparator. It uses an action conditioned diffusion transformer to predict future egocentric observations and evaluates imagined trajectories for navigation. This demonstrates that large pixel generative world models can support flexible inference time planning, but also highlights the contrasting objective of the present work, compact feature prediction, explicit component attribution, and bounded local rollout rather than billion parameter visual generation [5].

Hierarchical methods address long horizons by separating strategic and tactical decisions. Director, for example, learns a manager that proposes latent goals and a worker that executes them, extending Dreamer style behaviour over longer tasks [6]. This supports the thesis hypothesis that local prediction and long horizon navigation should not be forced into one timescale. However, hierarchy does not remove the need to represent persistent place structure, and a learned recurrent state can be difficult to inspect for false loop closure, revisitation, or frontier coverage.

8.4 Joint Embedding Predictive Architectures

JEPA reframes prediction as estimating a target representation from a context representation rather than reconstructing the input. LeCun's broader proposal argues for non generative predictive world models and hierarchical representations as a route towards autonomous machine intelligence [7]. I-JEPA demonstrated the approach for image representation learning [8], while V-JEPA extended feature prediction to video [9]. V-JEPA 2 subsequently combined large scale self supervised video pretraining with action conditioned post training for robotic planning [10].

V-JEPA 2.1 further targets dense spatial and temporal features through a dense predictive loss and deep self supervision, and reports strong results in depth estimation, visual navigation, and robotic manipulation [11]. It is therefore a relevant pretrained representation baseline for the RGB to BEV bottleneck. A trained JEPA should be compared against strong general purpose dense video features.

A central engineering problem is representation collapse. Non contrastive methods require a mechanism that prevents all inputs mapping to a constant. VICReg directly regularises invariance,

per dimension variance, and covariance [12]. EMA target approaches use a slowly updated target encoder and stop gradient branch. LeWorldModel instead combines next embedding prediction with SIGReg, which encourages a well spread latent distribution and supports stable end to end JEPA training from pixels [13]. Year 1's implementations provide direct empirical experience with all three families: VICReg in TinyAutoJEPA and TinyQuadJEPA, EMA targets in TinyQuadJEPA-v2, and SIGReg in leWMQuad.

Recent work has shifted from demonstrating that JEPA style planning is possible to identifying the design choices that make it work. DINO-WM predicts future pretrained DINO features and performs action sequence optimisation in feature space [14]. This is a strong comparator because it tests whether project trained JEPA states provide value beyond a powerful fixed visual representation. Terver et al. systematically examine architecture, objective, context, action conditioning, proprioception, and planning optimisation in JEPA world models [15].

Temporal Straightening shows that non collapsed semantic features are not necessarily well conditioned for planning. Its curvature regulariser encourages locally straighter latent trajectories, making Euclidean distance a better proxy for trajectory distance and improving the conditioning of gradient based planning [16]. This is directly relevant to the Year 1 finding that anti collapse alone did not preserve useful spatial structure. Representation geometry must be evaluated for action ranking and rollout, rather than only through variance and prediction loss.

Predictive Training with Latent Imagination provides a direct quadruped comparator. It adds a one step JEPA style predictor and SIGReg to a recurrent Unitree Go2 navigation policy, but the predictor is used only during training and is discarded at deployment [17]. The policy also uses depth, proprioception, and an explicit representation of local obstacles. It therefore establishes that predictive auxiliary training can improve quadruped navigation, while leaving open the distinct question studied here, whether an RGB only egocentric view, inference time rollout, and persistent topological memory improve navigation under long horizon partial observability.

These works mean that neither latent feature prediction nor action conditioned JEPA planning is sufficient as a novelty claim. The thesis must instead show why its learned representation, memory interface, or multi timescale structure changes performance under partial observability.

The Year 1 evidence also cautions against treating the anti collapse objective as the only important choice. A model can be non collapsed and scene dependent while still failing to preserve occupied space semantics or support useful action ranking. Likewise, low teacher forced prediction error can coexist with weak free running rollouts. The current v3 programme therefore places representation qualification and physical semantics before aggregate navigation, and it measures gradient interaction between auxiliary semantics and predictive learning.

8.5 Topological Memory and Graph Based Navigation

Topological navigation represents environments as places and connections rather than dense metric grids. Semi parametric Topological Memory stores a non parametric graph whose nodes correspond to observations and uses a learned retrieval model for localisation [18]. Active Neural SLAM

combines learned exploration with an explicit map and planner [19]. These systems demonstrate that persistent spatial structure and algorithmic planning can substantially improve navigation in unseen environments.

Graph structured world models move closer to the present thesis. L3P learns latent landmarks and reachability edges for long horizon planning [20]. RECON combines learned latent goal models with non parametric topological memory for open world exploration [21]. These are direct novelty threats to any broad claim of “learned topological memory for navigation”. They also provide useful design precedents: learned place or reachability models can coexist with algorithmic graph maintenance and search.

The thesis’ narrower gap lies at the interface with action conditioned JEPA prediction. A graph over observed features provides persistent connectivity but does not necessarily predict the local future consequences of candidate control sequences. A flat JEPA can score local action rollouts but does not necessarily preserve place identity over a long episode. Paper A tests whether the two are complementary and whether the JEPA state is a better memory substrate than fixed pretrained features when the same graph procedure is used.

The learned versus engineered boundary must remain explicit. If node creation uses a fixed latent distance threshold, graph insertion is algorithmic even though the latent is learned. If observation likelihoods or transition confidence are hand specified, the belief update is hybrid rather than end to end learned. The scientific question is not invalidated by hybrid structure, but the claim must identify which component is responsible. Sensitivity analysis and learned alternatives are therefore part of the planned evaluation.

8.6 Belief over Place and Perceptual Aliasing

Many topological systems commit to a single retrieved node. Under severe aliasing, this can create a catastrophic false loop closure or place merge. A belief distribution over candidate nodes is more consistent with the POMDP framing. Observation similarity supplies only one likelihood term, executed actions and graph transitions can disambiguate candidates over time. For example, two corridor views may be visually similar, but only one candidate node may be connected to the junction observed after a left turn.

The proposed belief state is therefore updated using similarity between the current observation and stored places, compatibility with the actions just executed, graph connectivity, and optional proprioceptive evidence. High entropy means that confidence remains spread across several plausible locations rather than concentrated on one. The planner can then select actions that help resolve the ambiguity, retain multiple candidate routes, or move more cautiously rather than prematurely committing to the single currently most likely location, the maximum a posteriori (MAP) estimate. Comparing the full belief distribution, the top- k most likely locations, and a MAP only estimate is necessary to determine whether retaining multiple hypotheses provides measurable navigation value.

A visible beacon may strongly disambiguate place and make memory appear better than it is in

beacon free regions. Evaluation must therefore separate pre sighting exploration, localisation where no beacon is visible, and post discovery navigation. If multiple beacons are used, identity association becomes an additional memory problem rather than a free landmark signal.

8.7 Planning: Reactive Costs, Latent Rollout, and Graph Search

A learned cost head can score the current state or immediate candidate action without simulating future latents. Such a reactive planner may be sufficient for collision avoidance and visible goal approach. A world model is only justified if rolling out predicted action consequences improves ranking, progress, or safety beyond this baseline. Paper A therefore includes a no rollout reactive controller with comparable heads and memory.

Local rollout and global graph search operate at different resolutions. CEM or a related optimiser can evaluate continuous command sequences over a short horizon. The graph can select a distant frontier, remembered beacon, or route through known places. The graph provides a subgoal or directional constraint, and the local world model decides how to execute safely in the current geometry. This avoids forcing a long autoregressive latent rollout across an entire maze, where compounding error is likely to dominate.

The architecture also permits diagnostic separation. Local action ranking accuracy measures whether predicted futures support control. Place and edge metrics measure memory. Coverage and task success measure the complete planner. A result in which memory improves coverage but rollout does not beat reactive scoring would be scientifically different from one in which both contribute. The planned ablations are designed to support this level of interpretation.

8.8 Hierarchical Predictive Modelling

The longer term H-JEPA direction extends this separation across timescales. A tactical predictor models short action blocks and immediate geometric consequences. A strategic predictor models slower transitions between place states, frontiers, rooms, or macro waypoints. This is related to hierarchical world models such as Director but remains distinct in its use of joint embedding prediction and explicit persistent place memory.

Recent work makes this hypothesis more specific. Hierarchical Planning with Latent World Models learns predictors at multiple temporal scales in a shared latent space and uses long horizon predictions as subgoals for short horizon control [22]. FF-JEPA similarly combines an action free latent planner with an action conditioned forward model to decompose long tasks into local optimisation problems [23]. These systems show that temporal hierarchy and latent subgoals are not sufficient novelty in isolation.

Mind the Gap evaluates a hierarchical extension of LeWorldModel and shows that hierarchy can fail when inference time high level search proposes latent subgoals that are poorly matched to the distribution understood by the frozen low level controller. Constraining search around macro actions encoded from training trajectories recovers useful hierarchical regimes [24]. The planned H-JEPA

evaluation must therefore measure subgoal executability and distribution compatibility, in addition to effective planning horizon, while controlling for memory, model capacity, and planning compute.

The hierarchy should not be assumed beneficial merely because the task is long horizon. A strong flat predictor plus graph memory may already provide sufficient temporal decomposition. The correct experiment therefore holds the encoder and memory substrate as constant where possible and tests whether a strategic predictor improves planning horizon, replan stability, sample efficiency, or transfer. Shared versus independent encoders and gradient routing are also empirical questions, the earlier eighty four million to one imbalance demonstrates that multi objective architectures can fail through optimisation even when their conceptual decomposition is reasonable.

8.9 Safety, Uncertainty, and Assurance Evidence

World model prediction errors can expose deviations between expected and observed dynamics. LeWorldModel reports surprise based detection of implausible events [13], but prediction residual is not automatically calibrated uncertainty. For safety use, the residual must be evaluated against concrete hazards and distribution shifts, including its detection lead time, false alarm rate, and effect on task performance. The same applies to memory entropy, graph inconsistency, repeated replanning, and safety vetoes, these are candidate evidence signals rather than guarantees of safe behaviour.

The distinction between a diagnostic and safety evidence is important. A measurable signal becomes relevant to an assurance argument only when it is linked to a defined hazard or safety requirement, evaluated within a specified operating context, and associated with an effective system response. A low latent prediction error, for example, does not by itself demonstrate safe navigation. It becomes useful evidence only if the prediction error has been shown to relate reliably to failures such as incorrect action ranking, insufficient clearance, collision risk, or loss of control within the planner's operating horizon.

The research therefore structures safety evidence around component boundaries. Perception gates test whether the learned spatial state preserves free and occupied space before navigation is attempted. Predictor tests measure whether action conditioned rollouts remain valid over the planning horizon. Memory tests quantify false place merges, false loop closures, unresolved localisation ambiguity, and recovery from incorrect hypotheses. Planner tests consider both collision avoidance and task capability, including coverage, progress, and completion. This decomposition does not make the system intrinsically safe, but it allows claims, assumptions, failure modes, and supporting evidence to be associated with the component or interaction to which they apply.

An assurance case provides a structured, context-bounded argument explaining why available evidence supports a claim that a system is acceptably safe. Goal Structuring Notation (GSN) makes the claims, argument strategies, context, assumptions, justifications, and evidence explicit [25]. Assurance argument patterns provide reusable structures for such reasoning, but they must still be instantiated using the hazards, requirements, operating conditions, and evidence of the particular system. A collection of strong experimental results is therefore not yet an assurance case unless the reasoning that connects those results to the safety claim is also made explicit.

8.10 Safety Assurance Frameworks and Evidence Mapping

Safety Assurance of Autonomous Systems in Complex Environments (SACE) and Assurance of Machine Learning for use in Autonomous Systems (AMLAS) provide complementary structures for organising this evidence. SACE addresses the autonomous system as a whole and considers its interaction with a complex environment. Its eight stages cover operating context assurance, hazardous scenario identification, the safe operating concept, safety requirements assurance, design assurance, hazardous failure management, out of context operation, and system verification [26]. AMLAS addresses an individual machine learned component within that wider system. Its six stages cover ML safety assurance scoping, ML safety requirements assurance, data management, model learning, model verification, and model deployment [27].

This distinction is important for the proposed architecture. The RGB to BEV encoder and decoder, action conditioned predictor, and learned risk or task heads are candidate ML components for which AMLAS style arguments may be developed. Any learned place similarity, observation likelihood, or transition confidence model may require a further ML argument. In contrast, graph insertion and search rules, the model predictive controller, the low level locomotion controller, and ROS 2 integration belong to the wider system design and must be addressed through a SACE style system argument. The topological memory is therefore a hybrid boundary: learned representations may be assured using AMLAS, while the algorithmic graph structure and its system level consequences remain part of the wider autonomous system assurance case.

SACE provides the outer argument by identifying the operating context, hazardous scenarios, and system level safety requirements. For the current quadruped case study, a provisional operating context would include bounded indoor environments, a specified range of maze geometry and appearance, and defined camera and robot configurations. Relevant hazardous scenarios include collision or fall following incorrect free space prediction, navigation to an unsafe region following a false place merge, repeated oscillation or vetoes that prevent task completion, and continued autonomous operation after the system has left its calibrated operating envelope.

The safe operating concept must address both operational harm and safety related non performance. Requirements may include avoiding contact above a defined severity, maintaining minimum clearance where practicable, detecting unreliable localisation or prediction before an unsafe action is executed, and entering a bounded fallback state when the operating assumptions no longer hold. Where failure to inspect an area could leave a hazard undetected, minimum coverage or task progress requirements may also form part of the safety argument. These requirements must be derived from the system hazard analysis and intended application rather than selected solely because they are convenient model metrics.

AMLAS then provides a structure for arguing that the learned components satisfy the safety requirements allocated to them. The proposed project artefacts map onto that structure as follows:

1. **Safety assurance scope and ML safety requirements.** The learned component boundaries, RGB input, BEV output, prediction horizon, and intended operating conditions define the scope of each ML component. System level requirements must then be translated into measurable ML

level requirements. Candidate requirements include minimum occupied space recall, maximum false free space prediction, preservation of free versus occupied training examples, action conditioned sensitivity, and acceptable degradation over a specified rollout horizon. The current gates provide candidate thresholds, but a complete assurance argument must justify how those thresholds follow from the relevant hazards and system level safety requirements.

2. **Data management assurance.** Scene level data separation, episode safe temporal sampling, and data generation manifests provide evidence for the AMLAS data argument. wrong RGB inputs help demonstrate that the model depends on the intended sensor input, while oracle tests help show that a proposed evaluation gate is achievable. Neither result alone demonstrates safety, they support the validity of the data and evaluation assumptions on which later evidence depends.
3. **Model learning assurance.** Source hashes, immutable attempt identities, model development logs, gradient diagnostics, and no regression criteria document how the model was produced. These artefacts are relevant to the earlier gradient starvation failure, because they show whether the predictive objective received a meaningful optimisation signal and whether joint training preserved previously qualified quality gates. Component specific baselines and ablations also help justify that an observed capability is attributable to the learned component instead of a classical scaffold.
4. **Model verification assurance.** Verification evidence includes occupied vs free space performance, one step and multi step prediction, action conditioned sensitivity, rollout degradation, and performance under defined distribution shifts. For learned memory components it also includes retrieval precision and recall, false place merges, and top- k localisation recall. Untouched and sealed evaluation, and per seed reporting support the claim that these results generalise within the specified context rather than reflecting a selected maze or checkpoint.
5. **System design and integrated verification.** Navigation level evidence belongs primarily to the SACE argument because it depends on interactions between learned and non learned components. Coverage, beacon capture, task completion, collision rate, path efficiency, and oscillation test whether the integrated design satisfies both physical and task related requirements. The planned ablations support design assurance by separating the contributions of perception, prediction, learned costs, memory, belief representation, graph search, and classical control.
6. **Deployment and out of context operation.** ROS 2 interface tests, camera and transform validation, command timing, hardware specific latency, simulator to physical comparisons, emergency stop testing, and complete physical replay logs provide deployment evidence. Prediction surprise, graph inconsistency, and repeated safety vetoes are candidate signals for recognising that the system is approaching or has crossed its assured operating boundary. Their usefulness must be demonstrated through false alarm rate, and the task cost of intervention. A fallback that prevents all movement may reduce collision exposure but would not satisfy a wider requirement for effective search or inspection.

SACE additionally uses assurance claim points to identify places where confidence in a claim or

artefact requires its own supporting argument [26]. This distinction maps directly onto the project's experimental custody controls. For example, an occupied space recall result is direct evidence concerning a perception requirement. The split manifest, one way access record, source hash, gate receipt, per seed interval, and independent review provide confidence that the result is relevant, unbiased, reproducible, and correctly attributed. Similarly, an oracle pass demonstrates that the evaluation criterion is technically achievable, but does not demonstrate that the learned model satisfies it.

The current programme therefore produces many artefacts that could populate AMLAS and SACE argument patterns, but it does not yet constitute a complete safety case. Important remaining work includes formalising the operational domain model, identifying and validating hazardous scenarios, deriving quantitative safety and ML requirements from those scenarios, defining stakeholder appropriate acceptance criteria, and specifying the minimum risk response outside the calibrated envelope. Organisational independence of verification must also be distinguished from merely keeping verification data hidden from the development process.

A further limitation is that AMLAS was developed primarily for offline supervised learning components used in perception and prediction. The proposed JEPA is substantially self supervised, and the memory system combines learned representations with online algorithmic state construction. AMLAS can therefore provide a valuable assurance structure, but direct claims of AMLAS conformance would require explicit tailoring and review [27]. The thesis contribution is not a replacement for AMLAS or SACE. It is an empirical investigation of whether physically grounded representation tests, bounded rollout evidence, belief diagnostics, planning outcomes, and runtime intervention traces can credibly instantiate parts of those established assurance structures for learned navigation.

8.11 Research Gap and Thesis Position

The literature supports the following conclusions:

- latent and JEPA style world models can support efficient planning, so action conditioned feature prediction alone is not novel
- topological and non parametric memories can support navigation in unseen environments, so graph memory alone is not novel.
- recent hierarchical latent planners already decompose control across temporal scales and generate latent subgoals, so a two level JEPA alone is not sufficient as a novelty claim [22–24]
- JEPA style predictive auxiliary training has already been transferred to a physical Unitree Go2, so predictive learning or quadruped deployment alone is not sufficient [17]
- within the literature reviewed up to August 2026, the remaining gap is the controlled interface between a physically grounded RGB only predictive state, inference time short horizon rollout, persistent place memory, and controlled quadruped navigation under strict held out evaluation.

The research therefore advances a staged and deliberately narrower position. First, it identifies what

an RGB only predictive representation must preserve before navigation claims are credible. Second, it evaluates the complementary roles of inference time local JEPA rollout and global topological memory under perceptual aliasing. Third, it tests whether a multi timescale JEPA adds value beyond that strong flat plus memory baseline, including whether strategic subgoals remain executable by the tactical controller.

Existing assurance frameworks provide the structure of the safety argument rather than the project specific evidence. The thesis therefore evaluates whether its perception gates, rollout tests, belief diagnostics, planning outcomes, and runtime intervention traces are sufficiently valid, calibrated, and traceable to instantiate parts of AMLAS and SACE style arguments for a bounded learned navigation system.

8.12 Search Terms and Review Maintenance

The literature search is maintained using combinations of the following terms:

- “joint embedding predictive architecture, “JEPA world model, “action conditioned JEPA, “V-JEPA 2-AC, and “DINO-WM.
- “latent world model planning, “feature-space planning, “model predictive control latent dynamics, and “autoregressive rollout error.
- “topological memory navigation, “latent landmark graph, “neural map, “active neural SLAM, and “place recognition perceptual aliasing.
- “POMDP belief navigation, “multi-hypothesis localisation, “false loop closure, and “state abstraction bisimulation.
- “hierarchical world model, “hierarchical JEPA, “latent subgoal planning, and “multi timescale prediction.
- “world model uncertainty, “prediction surprise OOD, “safe model predictive control, and “robotics safety case evidence.
- “V-JEPA”, “latent planning”, “hierarchical latent world model”, “latent goal”, “JEPA”, “predictive quadruped navigation”, and “predictive supervision”.

The review will be updated before each paper submission, with particular attention to new action conditioned JEPA planning work, memory augmented world models, and quadruped navigation benchmarks.

9 Progress to Date

9.1 Overview of the Experimental Progression

The work completed to date is best understood as an empirical progression through increasingly realistic predictive control systems. Each repository was used to answer a narrower question and expose assumptions that were hidden by the preceding system.

Repository	Primary question	Outcome and resulting decision
TinyAutoJEPA	Can a compact latent model support action conditioned prediction and model predictive control?	A VICReg encoder and multi step predictor were trained in CarRacing-v3. Auxiliary probes showed task relevant latent structure, and latent space MPC was viable, although brittle on difficult track geometry. This established the basic training and planning pipeline.
TinyQuadJEPA	Does the approach transfer from a 2D car to a legged embodiment?	A vision/proprioceptive encoder, action conditioned recurrent predictor, CEM planner, and frozen PPO locomotion controller were integrated in Genesis. Open floor waypoint behaviour was possible, but obstacle generalisation was poor.
TinyQuadJEPA-v2	Does a canonical EMA-target JEPA improve obstacle aware prediction?	Procedural walls, corridors, dead ends, textures, an EMA target encoder, and learned rollout costs were added. obstacle aware latent planning became feasible, but training stability and the integrity of the data generation pipeline became dominant concerns.
leWMQuad	Can end to end JEPA training be stabilised without an EMA target?	A ViT Tiny encoder, six layer action conditioned Transformer predictor, and SIGReg objective were adapted from LeWorldModel. Stable end to end training was achieved on the quadruped dataset, reducing the anti collapse objective to prediction plus a single distributional regulariser.

Repository	Primary question	Outcome and resulting decision
LeWMQuad-v2	Why does a trained latent planner fail on enclosed, long horizon navigation?	A camera clipping bug was identified and corrected. Executed action blocks, safer rendering, beacon labels, learned planning heads, online mapping, and extensive closed loop diagnostics were added. One familiar maze family was completed in closed loop, but the result retained a hand designed standoff route scaffold and tuned safety/target heuristics. Multi Scene evaluation then showed that the pooled visual latent did not encode transferable wall geometry and that coverage remained weak. This motivated explicit persistent memory and a spatially structured learned representation. SIGReg demonstrated to degrade spacial structure in the latent representations.
LeWMQuad-v3	Can an RGB only representation pass a physical perception gate before full JEPA and navigation evaluation?	The repository introduced fixed dataset roles, immutable attempts, preregistered gates, direct RGB to BEV learning, and controlled hypothesis testing. No current checkpoint has passed the complete perception qualification boundary.

9.2 Year 1 Findings from the Earlier Quadruped Stack

The earlier Mini Pupper and LeWMQuad experiments established several positive results. A frozen low level locomotion policy could be cleanly separated from a learned high level predictive planner. The action conditioned model learned non trivial scene dependence, and local predicted costs could support Open floor waypoint following, slalom like behaviour, beacon approach, and limited obstacle avoidance. The ViT-Tiny plus Transformer LeWM formulation also showed that an end to end pixel to latent world model could be trained on the collected trajectories without a target encoder when SIGReg was used as the anti collapse mechanism, though with some influences on latent structure which will be investigated as future work.

9.2.1 Retrospective on the strongest closed loop demonstration

The strongest maze family demonstration was a genuine success, but it was not yet a clean end to end learned navigation result that generalises on all maze types. Learned RGB/JEPA perception, persistent memory, and action scoring all contributed to the run. However, exploration still depended on tuned safety and target heuristics, which failed to generalise on other maze families. The result

therefore demonstrated that the learned components could be integrated into a functioning controller, not that they had independently learned a transferable navigation policy.

Later scaffold free learned policies could make progress on the same reference maze family. That maze had nevertheless been used repeatedly for training, debugging, threshold selection, and policy iteration. Its apparent success could therefore reflect maze specific visual and layout regularities rather than a general navigation rule. This possibility was borne out by broader evaluation.

The central state used by this stack was a pooled 96 dimensional JEPA image vector. It retained useful target and appearance information, but did not reliably preserve local wall geometry. On heldout data, the learned outcome head was anti predictive rather than weak, with an area under the receiver operating characteristic curve (AUC) of 0.41, the clearance head was close to chance at AUC 0.52. Collecting wall adjacent DAGger trajectories with exact labels did not remove the gap. Training accuracy reached 0.91 and 0.99 for the two heads, while unseen scene accuracy fell to 0.58 and 0.61. This isolated representation transfer, rather than an absence of labelled wall interactions, as the principal limitation.

closed loop control amplified these small geometric errors. At a permissive threshold the robot moved, but repeatedly drove into or pressed against walls. At a more conservative threshold, the safety guard vetoed almost every forward action and the robot became effectively stationary. This is a concrete example of why collision avoidance alone is not a sufficient safety result, as a controller can become nominally safe by suppressing the behaviour required to complete the task.

Exploration compounded perception failure. Approximately 72% of issued actions were turns, only 8–38 maze cells were visited within 700 control ticks, and many runs never brought most beacons into view. Perception and coverage therefore formed a coupled failure mode: inaccurate local geometry inhibited forward motion, while weak coverage prevented the agent from collecting the observations needed to find targets and recover from local errors.

The limitation was not confined to the formally held out maze. Across thirteen nominal training maze families, the full stack claimed only three of fifty two possible beacons. The successful reference maze family run was consequently a heavily tuned exception rather than evidence of broad in distribution competence. The concise conclusion is that the old stack learned enough appearance information and maze specific behaviour to complete one familiar environment family, but its global visual latent did not encode transferable spatial geometry. The remaining route and safety scaffolding masked this weakness until multi maze family evaluation.

Some later engineering probes still yielded useful local evidence. A router calibration change converted one previously worst case scene from zero of four beacon claims to three of four, and a physical baseline batch completed twelve of twelve runs without runtime or collision integrity failures. These should be interpreted as component and execution checks, not as evidence that the navigation problem had been solved. Frontier and mixed exploration variants achieved only seven claims across twenty four runs, with coverage often limited to a small fraction of the navigable area and large numbers of wall guard vetoes suppressing useful movement.

The associated post mortem identified four recurring methodological hazards:

1. **Scene leakage.** Random frame level splits can place views from the same generated maze into both training and evaluation, producing an unrealistically easy “held out” set.
2. **Reset contamination.** Temporal windows and displacement targets can cross episode resets unless episode boundaries are enforced at the dataloader level.
3. **Teacher forcing confusion.** One step or teacher forced prediction can look strong while autoregressive planning degrades rapidly at inference.
4. **Component misattribution.** A classical occupancy, frontier, depth, route, or safety component can dominate a hybrid system while aggregate success is attributed to the learned world model.

These findings directly informed the v3 implementation, and the Paper A ablation plan. Every learned contribution must now be isolated against a meaningful no component baseline, and every promotion decision must be made before access to untouched or sealed evaluation roles.

9.3 Current LeWMQuad-v3 Programme

9.3.1 Repository goal and evaluation custody

The repository goal is a fully learned, RGB only high level navigation stack for Unitree Go2 maze navigation. RGB observations feed a learned encoder, the encoder produces a spatial representation suitable for collision free navigation, an action conditioned JEPA predicts the next representation, and a planner uses the learned state and dynamics to navigate. The final frozen stack must progress through ordered development and navigation gates before a one way evaluation on the sealed V4 benchmark.

The train, selection, calibration, untouched, and sealed roles are fixed. Raw data construction and role separation have passed independent integrity audits. An oracle has passed the physical perception gate, demonstrating that the gate is achievable rather than internally inconsistent. Much of the planner, frontier/A*, runner, and immutable result infrastructure already exists, and the sealed V4 benchmark remains unopened. There is currently no evidence that the raw dataset should be repeatedly rebuilt or refined.

9.3.2 Original camera evidence model

The original v3 representation predicted hand structured camera evidence. Rays, first hit depth, ground clearance, and a fixed rasterised occupancy map. A small $N = 320$ fit passed all twenty six training fit checks, proving that the architecture had enough capacity to fit a small training population. It failed to generalise to the broader population. The promoted joint Camera+JEPA run completed 8,000 updates but failed every complete physical scope, and no checkpoint qualified for later navigation or G2.

Gradient diagnostics showed that the norm of the camera loss gradients was approximately 84 million times greater than that of the JEPA loss gradients. Because both branches were subject to a single

global gradient clip, the camera gradients dominated the combined norm and determined the clipping coefficient. This reduced the already small JEPA gradients to effectively negligible values, starving the BEV and predictor pathway of learning signal. The two gradient vectors were approximately orthogonal rather than oppositely aligned, indicating severe scale imbalance rather than strong objective conflict. Consequently, the experiment did not establish that the predictor architecture itself was inadequate, since the predictor had not received a meaningful optimization signal. This motivated staged component qualification and separate gradient normalization or clipping boundaries for the camera and JEPA objectives.

Camera only adaptation runs V1–V6 trained the visual encoder and camera evidence head while freezing the predictor path. Direct visual metrics and wrong RGB input improved, confirming that the model used image information. The final V6 run nevertheless passed zero of nine complete physical scopes. occupied space recall, raster quality, and rough motion behaviour remained inadequate, and late run progress flattened. Longer schedules, minor learning rate changes, seed repetition, and promotion of the “closest” failure were therefore rejected. The camera evidence line was retired.

9.3.3 Direct learned BEV

The next architecture replaced hand structured ray evidence with direct RGB to BEV learning. A ViT encoder produces image tokens, and learned BEV queries decode them into a 64×64 spatial state. This line showed strong early spatial learning and meaningful RGB dependence. Early joint predictor experiments still showed weak action conditioned transition learning, so perception qualification was separated from full JEPA training.

Prototype based representations learned meaningful spatial structure, but struggled to maintain a reliable balance between unknown, traversable, and occupied regions. A successor model failed an initial RGB control, indicating that its output was not sufficiently tied to the camera observation, and was therefore stopped before a full training attempt was consumed. An additional class balanced loss improved semantic learning, but did not prevent the representation of occupied space from collapsing. The evidence therefore narrows the problem: the encoder is learning genuine image dependent geometry, but does not yet preserve all navigation critical semantic classes reliably.

The run passed its update 100 gate. Its representation of spatial geometry, overall map classification loss, recognition of free space, and dependence on the RGB input continued to improve through update 400. However, it failed the preregistered requirements for recognising occupied space, maintaining a reasonable balance between free and occupied space predictions, and recovering the approximate extent of obstacles. The mechanism was therefore closed without a retry.

9.4 Current Scientific Position

The current bottleneck is perception qualification. Multiple wrong RGB controls show that the encoder is neither blind nor empty, and the direct BEV and signed distance experiments demonstrate learnable spatial structure. No complete encoder + BEV representation has yet passed the physical qualification boundary, especially on occupied space semantics and class balance. Consequently, no

checkpoint is authorised for predictor qualification, G2, closed loop navigation, runtime deployment, or held out use.

The experiment is perception pretraining and qualification, not full JEPA training. Training a predictor on a permanently frozen representation would produce a latent dynamics model, not the intended fully learned JEPA. A perception pass must therefore be followed by genuine joint learning in which the online encoder, BEV representation, and action conditioned predictor receive JEPA gradients against a stop gradient EMA target representation. The semantic objective should remain only as an auxiliary retention loss or no regression gate. JEPA and semantic gradients will require separate clipping, normalisation, parameter groups, or measured balancing to prevent a recurrence of the eighty four million to one starvation failure.

The repository is therefore closer to a trustworthy perception experiment than to a completed navigation stack, though we have demonstrated action conditioning and joint learning in prior experiments, incentivising the encoder to retain spatial understanding is the primary bottleneck. The existing navigation infrastructure should be reused once a representation has earned promotion, rather than redesigned in parallel with an unresolved perception bottleneck.

10 Research Methods, Tools, and Techniques

10.1 Software and Experimental Engineering

The primary method is construction and controlled evaluation of complete predictive navigation systems in Python and PyTorch. The repositories implement data collection, representation learning, action conditioned prediction, learned planning heads, model predictive control, online memory, evaluation, and audit tooling. Scientific hypotheses are introduced as narrow changes against a frozen baseline where possible. Unit, integration, wrong image, and mocked safety tests reject invalid mechanisms before expensive training.

The modelling stack includes convolutional and ViT encoders, learned query BEV decoders, recurrent and Transformer predictors, EMA target encoders, VICReg and SIGReg anti collapse objectives, physical and task heads, CEM style planning, frontier and graph search, and topological memory. A conventional low level locomotion controller is separated from high level navigation so gait performance is not confused with the learned world model.

10.2 Data Construction and Custody

Trajectory records include egocentric RGB, executed commands, episode identifiers, simulator pose, contact and clearance, and labels for planning heads. Collection policies cover teacher behaviour, frontier exploration, open loop motion, collision/recovery, and beacon approach. Temporal windows and displacement targets cannot cross episode boundaries.

Scene level train, selection, calibration, untouched, and sealed roles are fixed at construction time. Integrity is checked through manifests, raw frame audits, oracle evaluations, wrong image controls,

and small population fit tests. The oracle distinguishes an impossible gate from a failed learned mechanism.

10.3 Representation and Predictor Training

Representation learning proceeds through capacity, grounding, development generalisation, and physical scope gates. The current model maps ViT tokens to a 64×64 spatial state using learned BEV queries. If the representation qualifies, it becomes the initialisation for joint JEPA training. The online encoder, decoder, and predictor receive gradients against a stop gradient EMA target.

Training logs raw loss scales, pre and post control gradient norms, cosine similarity, clipping rates, and effective updates for each objective. Predictor evaluation separates one step teacher forced error from multi step error and measures action conditioned sensitivity, candidate action ranking, rollout degradation, and physical head consistency. Collapse diagnostics are necessary but are not substitutes for obstacle semantics or planning utility.

10.4 Memory, Belief, and Planning

The online memory is a graph whose nodes represent place observations or aggregated latent states and whose edges represent observed or predicted traversable transitions. Nodes may store visits, recency, frontier status, beacon evidence, and uncertainty. The implementation will state which insertion, merging, likelihood, and edge rules are learned or hybrid.

A belief b_t assigns confidence to candidate nodes and is updated from latent observation likelihood, action conditioned transition compatibility, graph consistency, and optional proprioception. The planner is evaluated using the full belief, a top- k approximation, and a MAP node. Memory measures include retrieval precision/recall, false merges, false loop closures, graph fragmentation, top- k recall, belief convergence, frontier quality, and recovery from a wrong hypothesis.

Local planning evaluates body frame action blocks through predicted latent futures and learned costs. Global graph search selects a frontier, remembered beacon, or intermediate place target. A reactive no rollout baseline tests whether prediction contributes beyond current state heads.

10.5 Evaluation and Statistical Analysis

Navigation outcomes include completion, beacon capture, coverage, novel area discovery, path efficiency, contact/collision rate, repeated vetoes, and oscillation. Results are split by beacon visibility. Compute measurements separate encoder, rollout, memory update, belief update, graph search, and total planning latency.

Competing systems will be evaluated on the same procedurally generated mazes, allowing performance to be compared maze by maze rather than across different test sets. Individual results, differences in coverage and beacon collection, and intervals showing the uncertainty around the overall difference will be reported. Statistical methods suitable for skewed, bounded, and success or

failure outcomes will be chosen in advance. The main outcome measures and diagnostic groups will be fixed before the untouched evaluation is opened. The final sealed benchmark will be used once to test the completely frozen system and will not be used to guide further model selection or tuning.

(***This is a significant weak point for me, ive done very little statistical analysis, let me know if the above is sensible or not***)

10.6 Physical Methods and Training Needs

ROS 2 will provide camera, transform, command, logging, and safety interfaces for the Unitree Go2. Physical runs use conservative speeds, a controlled arena, padded obstacles, an operator, and an emergency stop. Observations, commands, predictions, risk estimates, memory, belief, route choice, and interventions are logged.

Year 2 training priorities are ROS 2 and safe Go2 operation, probabilistic filtering under perceptual aliasing, latent geometry and state abstraction, multi timescale predictive modelling, and construction of traceable safety arguments.

11 Planned Outputs and Thesis Arc

The thesis is structured around a sequence of papers or paper scale contributions rather than a single monolithic system. (below is indicative, and may well not be realistic)

Stage	Provisional output	Contribution and dependency
Paper A	<i>An Action-Conditioned JEPA with Online Topological Memory for Quadruped Navigation</i>	Qualify an RGB only spatial predictive state, evaluate local action conditioned JEPA rollout, learned planning heads, online topological memory, and belief over place under perceptual aliasing. The primary comparison must show that any success transfers across novel mazes and is not carried by route or safety scaffolding. This is the current priority.
Paper B	Hierarchical JEPA for long horizon embodied navigation	Add tactical and strategic prediction timescales to the strongest Paper A baseline. Test whether hierarchy adds value beyond persistent memory and a strong flat predictor.
Paper C	Pretraining, robustness, and sim to real transfer	Compare from scratch task specific training with pretrained visual or video representations, evaluate transfer across scene, appearance, embodiment, and physical Unitree Go2 deployment.

Stage	Provisional output	Contribution and dependency
Paper D / thesis synthesis	Safety evidence for predictive navigation systems	Evaluate calibrated prediction, memory, and planning monitors, assemble failure maps, intervention evidence, and a structured argument over the operating envelope of the learned stack.

The sequencing is deliberately conservative. Paper A does not require H-JEPA to be implemented, and the H-JEPA paper does not assume that hierarchy will outperform a flat plus memory baseline. Physical deployment is not used as a substitute for sufficiently sophisticated simulation evaluation. The final thesis contribution is the accumulated evidence about what predictive representations, persistent memory, temporal hierarchy, and safety instrumentation each contribute to long horizon embodied navigation.

A Detailed LeWMQuad-v3 Status as of July 2026 - this can probably be commented out?

This appendix provides a concise technical record of the active repository position. It is included to make the relationship between completed experiments and future work unambiguous.

A.1 LeWMQuad-v2 retrospective

- The strongest one maze family result was a real closed loop demonstration using learned RGB/JEPA perception, memory, and action scoring, but exploration retained a hand designed standoff route scaffold and tuned safety/target heuristics.
- The reference maze family was repeatedly used during training, debugging, threshold selection, and policy iteration. Later scaffold free progress on that maze was therefore not valid evidence of transfer.
- The state was a pooled 96 dimensional JEPA image vector. On held out data, the outcome head achieved AUC 0.41 and the clearance head AUC 0.52, indicating that the latent did not reliably preserve local wall geometry.
- Wall adjacent DAgger data with exact labels produced training accuracies of 0.91/0.99 but unseen scene accuracies of 0.58/0.61. This implicated representation transfer rather than label scarcity.
- Permissive safety settings allowed motion but caused repeated wall contact, conservative settings vetoed almost all forward motion and effectively froze the robot.
- Approximately 72% of actions were turns, only 8-38 cells were visited in 700 ticks, and many runs did not observe most beacons.

- Across thirteen nominal training mazes, the stack claimed three of fifty two possible beacons. The one maze success was therefore a tuned exception, not broad in distribution competence.

The v2 stack learned useful appearance and target information and demonstrated genuine closed loop integration, but its pooled visual latent did not encode transferable spatial geometry. Route and safety scaffolding obscured this limitation until multi maze evaluation.

A.2 Established foundations

- The goal is a fully learned, RGB only high level navigation stack for Unitree Go2 maze navigation.
- Train, selection, calibration, untouched, and sealed roles are fixed and have passed role integrity audits.
- An oracle has passed the physical perception gate.
- Planner, frontier/A*, runner, custody, and immutable result infrastructure largely exist and should be reused.
- The V4 sealed benchmark remains unopened.
- There is no current evidence that the raw dataset requires rebuilding.

A.3 Closed architecture lines

- The ray/depth/clearance/fixed raster camera evidence model fit a small $N = 320$ training population but did not generalise.
- The promoted joint Camera+JEPA run completed 8,000 updates and failed every complete physical scope.
- Camera gradients were approximately 84 million times larger than JEPA gradients under a shared clip, gradients were approximately orthogonal, indicating starvation rather than proven architectural antagonism.
- Camera only V1-V6 improved direct visual metrics and passed wrong RGB input, but V6 passed 0/9 complete physical scopes.
- A hard Boolean/MAP raster adapter worsened aggregate raster performance.
- Seed repetition, schedule extension, promotion of the closest failure, and predictor training behind an unqualified representation were rejected.

A.4 Direct BEV evidence

- RGB is encoded into ViT tokens and decoded by learned BEV queries into a 64×64 spatial state.
- Direct BEV runs showed early spatial learning and meaningful RGB dependence.

- Prototype representations struggled with unknown/free/occupied balance.
- A multiprototype successor failed its update zero grounding control and was closed without training.
- A finalclass semantic anchor improved some semantic metrics but did not resolve occupied space collapse.
- The two field signed boundary model passed its update 100 gate and continued improving geometry, raster NLL, free recall, and grounding through update 400.
- It failed occupied recall, free/occupied balance, and rough occupied space criteria and was closed without retry.

A.5 Active semantic anchor experiment

A.6 Promotion logic

If V2 passes, the project will:

1. freeze the perception initialisation,
2. preregister genuine joint JEPA training,
3. train the online encoder, BEV state, and action predictor against a stop gradient EMA target,
4. preserve semantic performance through an auxiliary loss and no regression gates,
5. use separated gradient control,
6. select and calibrate only on fixed development roles,
7. proceed through the ordered navigation gates,
8. open V4 only after the complete freeze.

If V2 fails, the mechanism will be closed without retry. One materially different perception mechanism will be chosen from the failure signature while the RGB only, learned perception, genuine JEPA, and sealed maze constraints remain intact.

B Future H-JEPA Direction

B.1 Proposed three-tier stack

1. **System 1: low-level locomotion.** A conventional or factory controller maps bounded body-frame velocity commands to joint actions. It is frozen during high-level navigation experiments.

2. **Tactical JEPA.** A short-horizon predictor models the effects of action blocks on local geometric and physical state. It supports collision avoidance, traversability, visible-goal approach, and recovery.
3. **Strategic JEPA.** A slower predictor models transitions between place states, frontiers, rooms, or macro-waypoints. It proposes outcomes rather than raw joint-level controls.

The online topological memory remains a persistent substrate shared by the flat and hierarchical systems. This is important experimentally: hierarchy should be tested as an additional predictive mechanism, not introduced simultaneously with memory and then credited for the combined gain.

B.2 Key design questions

- Should tactical and strategic predictors share the complete encoder, share only an early trunk, or use independent encoders?
- How should gradients from the two temporal objectives be balanced and monitored?
- Should strategic actions be continuous relative waypoints, discrete graph transitions, frontier selections, or learned latent subgoals?
- Does the strategic predictor operate on pooled BEV states, topological node embeddings, or a separate representation?
- Is autoregressive finetuning required at both tiers?
- Which tier is responsible for uncertainty and intervention when predictions disagree?
- Does hierarchy improve effective horizon after controlling for memory, parameter count, and planning compute?

B.3 Required comparison

The principal baseline is the strongest flat action-conditioned JEPA plus online memory from Paper A. The H-JEPA will be considered successful only if it improves one of completion, coverage efficiency, long-route prediction, without unacceptable degradation prior benchmark figures. A null result would constrain the thesis by showing that explicit persistent memory supplies the required temporal abstraction without a second learned predictor.

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